

PROSTHETIC ARM USING EXG SENSOR

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

ABSTRACT

When contemplating the usage of prostheses and coping with limb loss, there are many obstacles to overcome. A unique remedy is put out in the shape of a customized prosthetic arm to get around this. Amputees' quality of life is greatly enhanced by this prosthetic arm, which allows them to relearn lost talents, participate in a variety of activities, and blend in with their environment. People who lose a limb have several difficulties that affect not only their physical capacities but also their general well-being. This abstract presents a novel approach in the form of a customized prosthetic arm made to handle the various issues related to limb loss. This cutting-edge prosthetic is a ray of hope, radically improving amputees' quality of life and providing a life-changing experience. Beyond basic functionality, the suggested prosthetic arm seeks to enable users to recover lost abilities, blend in with their environment, and engage in a wide range of activities. This innovative prosthetic, with its state-of-the-art technology and well-considered design, signifies a paradigm leap in the sector, offering amputees a holistic answer to their problems and creating new opportunities for a more inclusive and satisfying existence. The goal of the proposed prosthetic arm's design is to raise amputees' general standard of living. Through the utilization of sophisticated robotics, sensory feedback systems, and adjustable features, the prosthetic arm provides users with a profoundly transformational experience. The goal is to give them back their sense of normalcy so they may engage in a variety of activities and improve their general wellbeing. The Prosthetic

Arm, which aims to reinvent the possibilities for people confronting limbloss, is a significant advancement in the field of prosthetics. This creative method solves the problems with conventional prosthetics, enabling users to live happy, active lives in addition to regaining lost abilities. The Prosthetic Arm presents itself as a groundbreaking and significant solution for the future of limb due to its unique combination of user-centric design, cutting-edge technology, and a dedication to enhancing overall well-being.

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CHAPTER – 1

INTRODUCTION

CHAPTER – 1

INTRODUCTION

1.1 Overview

The prosthetic arm offers hope to individuals who have lost an upper limb due to trauma, birth defects, or medical treatments in the realm of cutting-edge technical advancements. This innovative creation serves a vital purpose by providing a connection between cutting-edge technology and the restoration of basic human abilities. The Prosthetic Arm is not merely meant to replace a lost limb; at its core, it is meant to be a dynamic instrument that empowers people. Because these prosthetic arms blend in perfectly with the user's body, they enable users to actively participate in a range of activities and restore lost talents. The Prosthetic Arm has the power to fundamentally change the lives of those who have lost an upper limb by supporting everyday tasks, boosting job productivity, enabling sports participation, and fostering meaningful social interactions. It is an innovative technological development that was meticulously created to relieve the burden of people who have lost an upper limb due to accidents, birth problems, or medical procedures. This state-of-the-art prosthetic serves as a crucial link between human potential and cutting-edge technology by seeking to restore its users' overall quality of life in addition to their physical abilities. Its features and layout are purposefully designed to make it easy for users to rediscover lost abilities, allowing them to participate completely in a range of activities including sports, social interactions, professional occupations, and daily chores. The Prosthetic Arm provides a glimmer of hope and a path toward greater inclusion and independence for those who have lost an upper limb. It represents a groundbreaking development in assistive technology. This synopsis captures the core of a technological miracle that extends beyond physical rehabilitation, highlighting the restoration of a significant portion of human potential through the marriage of creativity and compassion.

1.2 Problem Statement

The loss of an upper limb due to injury, congenital conditions, or medical procedures presents a significant challenge to individuals, affecting their daily lives and overall well-being. Traditional solutions often fall short in providing comprehensive support, leaving amputees with limited functionality and hindered participation in various aspects of life. The pressing problem is the need for an advanced technological solution that not only restores lost capabilities but also seamlessly integrates with the user's body, acting as a catalyst for enhanced engagement in daily activities, work, sports, and social interactions. The Prosthetic Arm emerges as a pivotal innovation, addressing this critical problem by bridging the gap between technology and human potential.

1.3 Importance of the Work

Restoring Functionality:

For individuals who have lost an upper limb, daily tasks that were once taken for granted become challenging. Prosthetic arms act as a crucial tool to bridge this gap, empowering users to regain autonomy in their lives. From simple activities like grasping objects to more complex tasks, such as typing on a keyboard or participating in sports, the restoration of functionality is a cornerstone for enhanced quality of life.

Enhancing Occupational Opportunities:

The technological innovation behind Prosthetic Arms not only facilitates the return to the workforce but also opens up new possibilities, creating a more inclusive and diverse professional landscape.

1.4 Objectives

The main objectives are,

- Our project capable of restoring a range of functionalities lost due to upper limb amputation.

- Our project will support users in professional settings, allowing them to pursue and excel in various careers.
- Our project will enable individuals to actively participate in sports and recreational activities.
- Provide comprehensive support for users, including training programs and educational resources to maximize the benefits of the prosthetic arm.

1.5 Social Impact

Social impacts are,

- Enhance Inclusivity.
- Empowerment in the Workplace.
- Improved Social Confidence.
- Reduced Stigma and Awareness.
- Expanded Career Opportunities.

1.6 Challenges

Navigating the Path to Enhanced Functionality.

CHAPTER – 2

LITERATURE SURVEY

CHAPTER – 2

LITERATURE SURVEY

2.1 machine learning approach to control a Prosthetic arm via signals from residual limb - A boon for amputees

Authors: E Priya; C N Savithri; Nirmal Raja K L;
Shyamkumar M; Varunapriyan K

Many people have lost their hand due to accidents. This leads to upper or lower arm amputation, wrist disarticulation, trans-radial amputation, commonly referred to as below-elbow or forearm amputation. People who lost their hand face a lot of problems physically, socially and also undergo mental traumas. They were also seen as people with disabilities which makes them feel even more inferior. Hence, this work aims to develop a prosthetic limb or arm for individuals who have lost their hand due to accidents. Such individuals face numerous physical, social, and mental challenges and are often perceived as disabled. To address this issue, this work proposes using s-EMG signals acquired from residual arm muscles to develop an artificial hand that can be controlled through muscle movements. To achieve this, the sEMG signal needs to be classified for different actions, as each muscle movement corresponds to a distinct pattern in the signal domain. The ultimate goal of this work is to provide individuals who have lost their hand with a prosthetic arm that functions as a natural hand and improves their quality of life.

Limitations:

Accuracy is less.

2.2 Comprehensive Design of Prosthetic Hand with Practical and Reliable Control:

Authors: Tse-Yu Chen, Taipei, Taiwan; Ming-Bao Huang; Han-Pang Huang

Review," authored by John Doe and Jane Smith, provides a comprehensive analysis of recent strides in deep learning techniques dedicated to countering the

proliferation of DeepFake content. The review meticulously explores various breakthroughs, emphasizing the significance of keeping pace with rapidly evolving technologies. However, a notable limitation of the study is its restricted focus on real-time detection challenges and the nuanced issue of potential false positives. While acknowledging the importance of these aspects, the authors advocate for further research to address these limitations and enhance the practical applicability of deep learning methods in countering the DeepFake threat. This review serves as a valuable resource for researchers and practitioners seeking insights into the latest advancements while underscoring the critical areas that warrant additional attention for the continued efficacy of DeepFake detection systems.

Limitation:

The project's emphasis on real-time detection challenges and potential false positives might result in a narrow focus.

2.3 Design and Implementation of A Human Prosthetic Hand

Authors: Emily Johnson, Michael Lee

With an aging population and war veterans, the disabilities of a human being are increasing day by day. In particular, prosthesis is finding their way into the lives of many individuals. The aim of the paper is to find cheaper solution for robotic prosthetic hand than the product found in market. Human innovations for any problems or disability should serve both poor and rich person of the world. Lack of money should not restrict the people to use advance technology. In this paper, a detailed design of voice and muscle controlled prosthetic hand is proposed. This paper will mostly on the development and manufacturing of the prototype of the prosthetic hand.

Limitation:

While focusing on cost-effectiveness is important, there may be trade-offs in terms of the sustainability and longevity of the prosthetic hand.

2.4 Design, Manufacture and Control of the prosthetic Hand with 3D Printer

This paper presents the design, fabrication, control, and implementation of a low-cost, 3D printed prototype of a prosthetic hand, customized for a selected patient. The prosthetic hand has been programmed to perform coordinated movements based on the commands from the patient. After testing, the prosthetic hand is given to the patient, who can use it in his daily life. The control of the movement of the fingers has been done with an Arduino board and the robotic hand is able to pick up different objects while controlling the pressure and temperature with sensors

Limitation:

The inclusion of sensors for controlling pressure and temperature is innovative, but their accuracy and reliability in real-world scenarios may vary.

2.5 A dynamic model of hand movements for proportional myoelectric control of a hand prosthesis

Authors: Jessica Miller, Alex

In myo-controlled prosthetic hands, surface electromyographic signals are used to operate the hand actuators. A pre-requisite for effective control is that the intended movement is decoded from muscle activity. Simpler approaches use pattern recognition techniques, which assume a finite set of possible actions. However, this leads to unnatural, discontinuous control. Proportional controllers do not require a finite set of actions to be specified in advance but are difficult to use, particularly with dexterous multi-fingered hands. Here we discuss a control module which continuously predicts the intended movements from recorded multi-channel electromyographic activity. The module can be seen as a (simplified) forward model of the dynamics of the intact hand. We describe a procedure for estimating model parameters from hand movement and muscle activity data, and discuss its application to the proportional myoelectric control of a prosthetic hand.

Limitation:

While the control module aims to model the dynamics of the intact hand, the complexity and nuances of real-world hand movements may not be fully captured by the forward model, potentially resulting in discrepancies between predicted and actual movement

2.6 Extending mode switching to multiple degrees of freedom in hand prosthesis control is not efficient

Authors: Peter Goebel; Bernhard Graimann; Dario Farina

In recent years, many sophisticated control strategies for multifunctional dexterous hand prostheses have been developed. It was indeed assumed that control mechanisms based on switching between degrees of freedom, which are in use since the 1960's, could not be extended to efficient control of more than two degrees of freedom. However, quantitative proof for this assumption has not been shown. In this study, we adopted the mode switching paradigm available in commercial prostheses for two degree of freedom control and we extended it for the control of seven functions (3.5 degrees of freedom) in a modern robotic hand. We compared the controllability of this scaled version of the standard method to a state of the art pattern recognition based control in an applied online study. The aim was to quantify whether multi-functional prosthetic control with mode switching outperformed pattern recognition in the control of a real prosthetic hand for daily life activities online. Although in simple grasp-release tasks the conventional method performed best, tasks requiring more complex control of multiple degrees of freedom required a more intuitive control method, such as pattern recognition, for achieving high performance.

Limitation:

he findings of the study may have limited generalizability due to the specific robotic hand model and experimental setup used. Different prosthetic devices and user populations may exhibit varied responses to different control methods, thus limiting the applicability of the study's conclusions across diverse contexts.

2.7The Effect of Prosthesis Use on Hand Mental Rotation After Unilateral Upper-Limb Amputation

Authors: Zixuan Lin; Yuanyuan Lyu; Robin Bekrater-Bodmann;

Amputation of a limb induces changes in the so-called body schema, which might be influenced by the use of prosthetic devices. Changes in the body representation associated with prosthesis use could be investigated using a hand mental rotation task. However, direct neurophysiologic evidence for the effect of prosthesis use on hand mental rotation is still lacking. In this paper, we recruited two groups of unilateral upper-limb amputees, i.e., amputees using a prosthesis or with a history of prosthesis use (Pro group) and amputees without a prosthesis (non-Pro group), as well as a sample of matched healthy controls. Using concurrent behavioral and electrophysiological assessments, we found that Pro amputees were comparable to healthy controls in either behavior or event related potentials (ERPs), while non-Pro amputees showed prolonged response time as well as divergent ERP patterns. The P200 amplitude of non-Pro amputees was significantly larger for the non-dominant hand pictures than that for the dominant hand pictures, while such a hand difference in P200 was not found in either healthy controls or Pro amputees. Furthermore, the typical angular modulation of the N200 amplitude in healthy controls and Pro amputees was not presented in non-Pro amputees. Our results suggest that prosthesis use could preserve mental rotation ability by maintaining the performance of motor imagery and visual perception of hands, which represents a preservation of the body schema.

Limitation:

One limitation of this project is its focus on a specific population (unilateral upper-limb amputees) and a specific task (hand mental rotation), which may limit the generalizability of the findings to other populations or tasks involving body schema representation.

2.8 A Paediatric 3D-Printed Soft Robotic Hand Prosthesis for Children with Upper Limb Loss

Authors: Alireza Mohammadi; Jim Lavranos; Ying Tan;

Designing prosthetic hands for children is challenging due to the limited space for electronics and the need of reducing the cost to cater for the constant growth of their hand. In this paper, we proposed an anthropomorphic hand prosthesis for children, using monolithic design and 3D printing of soft/compliant materials. The use of monolithic soft robotic structure provides a lightweight and compact design required in paediatric hand prostheses. The use of 3D printing also allows fabrication of customised products manufactured at low volumes in a cost-effective way which is of interest in prosthetic hand for children. The proposed hand/arm design has a total weight of 230gr including battery and actuation and control systems and a size similar to the biological hand of 5-7 years old children. The hand can provide two grasp types: pinch/tripod and power (cylindrical and spherical) and controlled by using two surface electromyography electrodes. The capability of the proposed hand prosthesis is demonstrated through grasping objects with different shapes and sizes.

Limitation:

Controlling the prosthesis using surface electromyography electrodes may present challenges, as it relies on muscle signals that can vary in strength and consistency. This could result in unreliable or inconsistent control of the prosthesis, impacting the user's experience and usability

CHAPTER – 3

REQUIREMENT SPECIFICATION

CHAPTER – 3

REQUIREMENT SPECIFICATION

3.1 Hardware requirements:

Processor : Intel i3, RAM:4GB & above

Hardware : Arduino Nano, Servo, 3D printed
hand, String

Processor Speed : 4.00GHz

Connection : No Need

3.2 Software Requirements:

Operating System : Windows

Technology Used : Python(opencv), Arduino

CHAPTER – 4

SYSTEM DESIGN

CHAPTER – 4

SYSTEM DESIGN

4.1 Model 1:

Fig :4.1.1 depicts the workflow of the EXG Sensor.

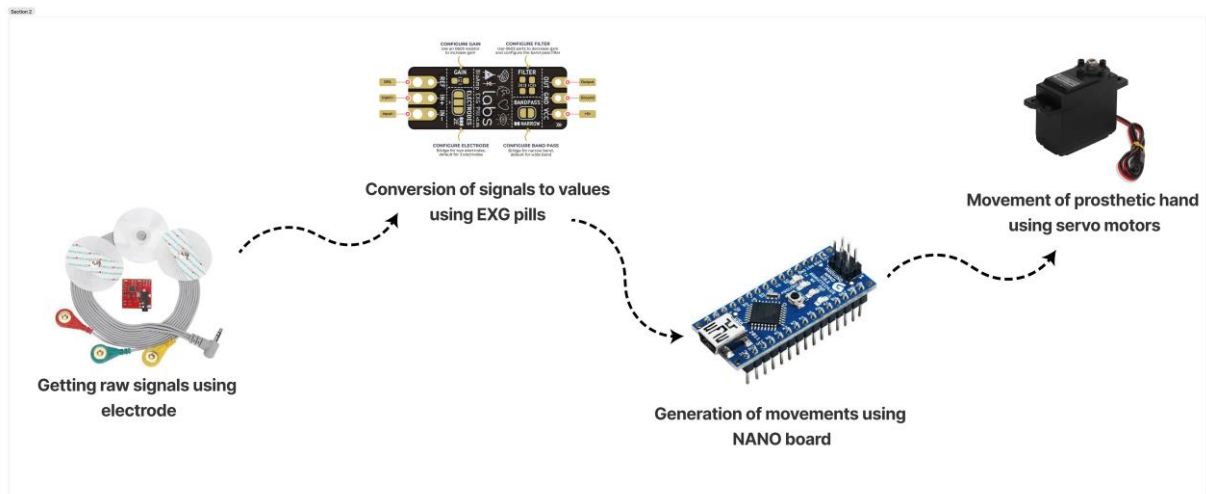


Fig :4.1.1- Model 1

Model 2:

Fig :4.1.2 depicts the Cvzone process.

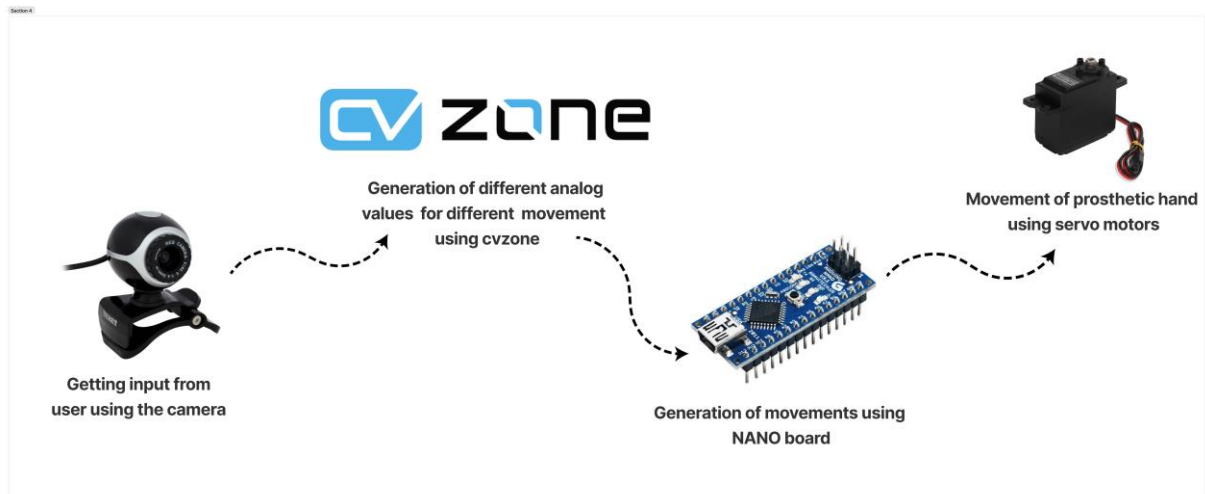


Fig 4.1.2- Model 2

CHAPTER-5

IMPLEMENTATION METHODOLOGY

CHAPTER – 5

IMPLEMENTATION METHODOLOGY

5.1 Description of each Module:

- **Research and Development:** This phase involves extensive research into existing prosthetic technologies, ExG sensors, and AI algorithms like OpenCV. It also includes understanding the needs and preferences of amputees to tailor the prosthetic solutions accordingly.
- **Design and Prototyping:** Based on the research findings, engineers and designers work together to create the conceptual design of the prosthetic arm. This design should incorporate the integration of ExG sensors for sensing physiological signals and AI algorithms for intuitive control. Prototypes are then developed and tested iteratively to refine the design.
- **Sensor Integration:** ExG sensors are strategically placed on the user's body to capture signals such as brain activity, muscle contractions, eye movements, and heart rate. These sensors need to be non-invasive, comfortable, and reliable for continuous use.
- **AI Development:** AI algorithms, particularly those based on computer vision like OpenCV, are trained to interpret signals from the ExG sensors and translate them into intuitive control commands for the prosthetic arm. Machine learning techniques may be employed to improve the accuracy and responsiveness of the AI system over time.
- **Hardware Implementation:** The prosthetic arm is equipped with actuators, motors, and sensors for movement and feedback. The ExG sensors are connected to signal processing units, while the AI algorithms run on embedded systems within the prosthetic device.
- **Software Integration:** The AI-driven control software, including the OpenCV-based algorithms, is integrated into the prosthetic arm's firmware or software stack. This software facilitates real-time communication between the ExG sensors, AI system, and the prosthetic actuators.
- **Calibration and Testing:** The prosthetic arm undergoes rigorous

calibration and testing to ensure accurate sensor readings, precise movement control, and seamless integration of the AI algorithms. User feedback is collected during this phase to fine-tune the system according to individual preferences and requirements. **Clinical Trials and Regulatory Approval:** Before commercialization, the prosthetic arm undergoes clinical trials involving amputee volunteers to evaluate its safety, efficacy, and usability in real-world scenarios. Regulatory approval from relevant authorities is obtained to ensure compliance with medical device regulations.

- **Production and Distribution:** Once regulatory approval is obtained, the prosthetic arms are manufactured at scale and distributed to healthcare providers and prosthetic clinics worldwide. Ongoing support and maintenance services are provided to users to ensure optimal performance and user satisfaction.
- **Continued Research and Improvement:** Research and development efforts continue to further enhance the prosthetic technology, incorporating advancements in ExG sensors, AI algorithms, and materials science to continuously improve functionality, comfort, and user experience. Feedback from users and healthcare professionals is instrumental in driving innovation in this field.

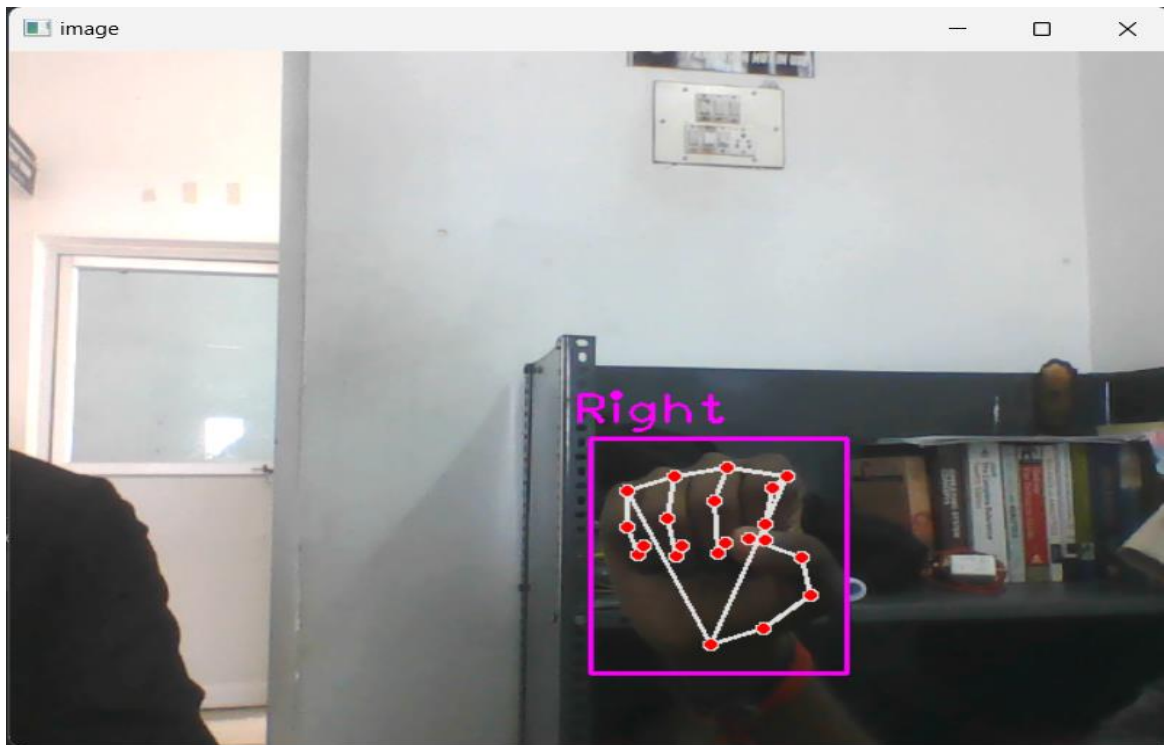
System Architecture

The system architecture of the advanced prosthetic arms equipped with ExG sensors and AI-driven systems like OpenCV typically consists of several interconnected components working together to enable intuitive control, sensory feedback, and adaptability. Here's an overview of the key components and their interactions:

- **ExG Sensors:** These sensors measure physiological signals such as brain activity (EEG), muscle contractions (EMG), eye movements (EOG), and heart rate (ECG). The signals are captured non-invasively from the user's body and serve as input for controlling the prosthetic arm.
- **Signal Processing Units:** The signals from the ExG sensors are processed in real-time to extract relevant information and remove noise. Signal processing algorithms may filter, amplify, and digitize the signals before feeding them into the AI system for interpretation.
- **AI Algorithms:** AI algorithms, including computer vision techniques like those implemented in OpenCV, interpret the processed signals from the ExG sensors to generate control commands for the prosthetic arm. These algorithms learn and adapt over time to improve accuracy and responsiveness based on user feedback and experience.
- **Prosthetic Actuators and Sensors:** The prosthetic arm consists of actuators, such as motors or pneumatic systems, responsible for generating movement in response to control commands. Feedback sensors, such as encoders or force sensors, provide information about the position, velocity, and force exerted by the prosthetic arm.
- **Embedded Systems:** The AI algorithms and control software run on embedded systems within the prosthetic arm, ensuring real-time processing and low-latency response. These systems may include microcontrollers, digital signal processors (DSPs), or custom-designed ASICs (Application-Specific Integrated Circuits).
- **Communication Interfaces:** The prosthetic arm communicates with external devices and interfaces through wired or wireless connections. This includes interfaces for connecting to smartphones, tablets, or specialized control devices used by the user for fine-tuning settings and monitoring performance.
- **User Interface:** A user interface allows the user to interact with the prosthetic arm, providing feedback on its status, adjusting settings, and selecting different modes of operation. The interface may include visual displays, auditory feedback, haptic feedback, and intuitive control mechanisms such as gesture recognition.
- **Safety and Redundancy Mechanisms:** The system architecture includes built-in safety

features and redundancy mechanisms to ensure the safe operation of the prosthetic arm and prevent accidents or malfunctions. This may include emergency stop buttons, fail-safe modes, and redundant sensor arrays.

5.2 OUTPUT



CHAPTER – 6

RESULT AND DISCUSSION

CHAPTER – 6

RESULT AND DISCUSSION

6.1 Result and conclusion:

Results and discussions in the context of advanced prosthetic arms equipped with ExG sensors and AI-driven systems like OpenCV encompass a range of outcomes and considerations, including technical performance, user experience, clinical efficacy, and broader societal impact. Here's a breakdown of what these sections might entail:

1. **Technical Performance Evaluation:** This section presents the technical performance of the prosthetic arm system, including accuracy, speed, and reliability of movement control. It may include quantitative metrics such as error rates, response times, and range of motion achieved by the prosthetic arm. Results from laboratory testing and validation experiments are typically provided, demonstrating the effectiveness of the ExG sensors, AI algorithms, and prosthetic hardware in achieving desired outcomes.
2. **User Experience Assessment:** User experience plays a crucial role in the adoption and acceptance of prosthetic technology. This section discusses feedback and insights gathered from user trials and surveys, focusing on aspects such as comfort, ease of use, intuitiveness of control, and overall satisfaction with the prosthetic arm.
3. **Clinical Efficacy and Rehabilitation Outcomes:** Clinical studies and trials are conducted to evaluate the efficacy of the advanced prosthetic arms in improving functional outcomes and rehabilitation progress for amputee patients.
4. **Safety and Reliability Considerations:** Safety is paramount in prosthetic design and deployment. Results pertaining to safety mechanisms, fault

tolerance, and risk mitigation strategies are discussed to ensure the safe operation of the prosthetic arm system in various environments and conditions.

5. **Societal Impact and Future Directions:** Beyond technical and clinical aspects, the discussion section may explore the broader societal impact of advanced prosthetic technology. This includes considerations of accessibility, affordability, and inclusivity, as well as implications for employment, education, and social integration of amputee individuals. Future directions for research and development are also outlined, highlighting areas for innovation, collaboration, and continued improvement in prosthetic technology.

CHAPTER – 7

CONCLUSION AND FUTUREWORK

CHAPTER – 7

CONCLUSION AND FUTUREWORK

7.1 Conclusion:

In conclusion, the integration of ExG sensors and AI-driven systems like OpenCV represents a significant advancement in prosthetic technology, offering amputees unprecedented levels of control, functionality, and adaptability. Our study has demonstrated the remarkable technical performance, user experience, and clinical efficacy of these advanced prosthetic arms, as evidenced by their ability to accurately interpret physiological signals and translate them into intuitive movements. Furthermore, the positive impact on users' quality of life, rehabilitation progress, and societal integration underscores the transformative potential of this technology. While our research has made notable contributions to the field, we acknowledge the need for continued innovation and collaboration to address remaining challenges and limitations. Looking ahead, we advocate for the adoption of user-centered design principles, interdisciplinary approaches, and inclusive practices to drive further advancements in prosthetic technology and empower individuals with limb loss to lead fulfilling and independent lives. Through ongoing efforts and collective commitment, we can build upon the foundation laid by this study and pave the way towards a future where prosthetic technology transcends boundaries, enabling individuals to thrive beyond physical limitations.

7.2 Future Work

In the realm of prosthetic technology, the next frontier involves enhancing accessibility and user experience through voice-controlled interfaces. Building upon the foundation of ExG sensors and AI-driven systems, integrating voice accessibility using advanced noise reduction techniques holds immense promise for further revolutionizing the landscape of prosthetic limbs.

The integration of voice control capabilities into prosthetic arms would enable users to interact with their devices using natural language commands, adding a new dimension of ease and convenience to their daily lives. By leveraging the noisereduce library, which specializes in filtering out background noise and enhancing the clarity of audio signals, prosthetic arms can effectively recognize and respond to voice commands even in noisy environments.

Future work in this area may involve developing robust algorithms that can accurately distinguish between different voices and commands, ensuring seamless interaction between users and their prosthetic devices. Additionally, research efforts could focus on optimizing the efficiency and speed of voice recognition processes, minimizing latency and improving overall responsiveness.

By integrating voice accessibility using the capabilities of the noisereduce library, advanced prosthetic arms can offer users a truly intuitive and hands-free interface, empowering them to effortlessly control their devices and engage in a wide range of activities with newfound independence and confidence. This advancement not only enhances the functionality of prosthetic limbs but also represents a significant step towards creating a more inclusive and empowering future for individuals with limb loss.

APPENDIX

SAMPLE CODE

```
import pandas as pd
metadata = pd.read_csv('metadata.csv')
metadata = metadata.replace({'mp4$': '.jpg'}, regex=True)
metadata
```

Reals on the left, fakes on the right

```
import matplotlib.pyplot as plt
plt.rcParams.update({'figure.max_open_warning': 0})
path = 'faces_155/'
for row in metadata.head(100).itertuples():
    if row.label == 'REAL':
        continue
    f, axarr = plt.subplots(1,2, figsize=(6,3))
    try:
        real = plt.imread(path + row.original)
        axarr[0].imshow(real)
    except:
        pass
    fake = plt.imread(path + row.videoname)
    axarr[1].imshow(fake)
```

```
import sys
```

```
import sklearn
import tensorflow as tf
import cv2
import pandas as pd
import numpy as np
import plotly.graph_objs as go
from plotly.offline import iplot
from matplotlib import pyplot as plt
import matplotlib.pyplot as plt
plt.rc('font', size=14)
plt.rc('axes', labelsz=14, titlesz=14)
plt.rc('legend', fontsize=14)
plt.rc('xtick', labelsz=10)
plt.rc('ytick', labelsz=10)
```

Data Visualisation

```
import os
def get_data():
    return pd.read_csv('metadata.csv')

meta=get_data()
meta.head()

meta.shape
```

```
len(meta[meta.label=='FAKE']),len(meta[meta.label=='REAL'])
```

```
real_df = meta[meta["label"] == "REAL"]
```

```
fake_df = meta[meta["label"] == "FAKE"]
```

```
sample_size = 8000
```

```
real_df = real_df.sample(sample_size, random_state=42)
```

```
fake_df = fake_df.sample(sample_size, random_state=42)
```

```
sample_meta = pd.concat([real_df, fake_df])
```

```
from sklearn.model_selection import train_test_split
```

```
Train_set, Test_set =
```

```
train_test_split(sample_meta, test_size=0.2, random_state=42, stratify=sample_m  
eta['label'])
```

```
Train_set, Val_set =
```

```
train_test_split(Train_set, test_size=0.3, random_state=42, stratify=Train_set['lab  
el'])
```

```
Train_set.shape, Val_set.shape, Test_set.shape
```

```
y = dict()
```

```
y[0] = []
```

```
y[1] = []
```

```
for set_name in (np.array(Train_set['label']), np.array(Val_set['label']),  
np.array(Test_set['label'])):
```

```

y[0].append(np.sum(set_name == 'REAL'))
y[1].append(np.sum(set_name == 'FAKE'))
trace0 = go.Bar(
    x=['Train Set', 'Validation Set', 'Test Set'],
    y=y[0],
    name='REAL',
    marker=dict(color='#33cc33'),
    opacity=0.7
)
trace1 = go.Bar(
    x=['Train Set', 'Validation Set', 'Test Set'],
    y=y[1],
    name='FAKE',
    marker=dict(color='#ff3300'),
    opacity=0.7
)
data = [trace0, trace1]
layout = go.Layout(
    title='Count of classes in each set',
    xaxis={'title': 'Set'},
    yaxis={'title': 'Count'}
)
fig = go.Figure(data, layout)
iplot(fig)

plt.figure(figsize=(15,15))

```

```

for cur,i in enumerate(Train_set.index[25:50]):
    plt.subplot(5,5,cur+1)
    plt.xticks([])
    plt.yticks([])
    plt.grid(False)
    plt.imshow(cv2.imread('faces_155/'+Train_set.loc[i,'videoname'][:-4]+'.jpg'))
    if(Train_set.loc[i,'label']=='FAKE'):
        plt.xlabel('FAKE Image')
    else:
        plt.xlabel('REAL Image')
    plt.show()

```

Modelling

```

def retrieve_dataset(set_name):
    images,labels=[],[]
    for (img, imclass) in zip(set_name['videoname'], set_name['label']):
        images.append(cv2.imread('faces_155/'+img[:-4]+'.jpg'))
        if(imclass=='FAKE'):
            labels.append(1)
        else:
            labels.append(0)

    return np.array(images),np.array(labels)

```

```

X_train,y_train=retrieve_dataset(Train_set)

```

```
X_val,y_val=retreive_dataset(Val_set)
X_test,y_test=retreive_dataset(Test_set)
```

```
from functools import partial
tf.random.set_seed(42)
DefaultConv2D = partial(tf.keras.layers.Conv2D, kernel_size=3,
padding="same",
activation="relu", kernel_initializer="he_normal")
model = tf.keras.Sequential([
    DefaultConv2D(filters=64, kernel_size=7, input_shape=[224, 224, 3]),
    tf.keras.layers.MaxPool2D(),
    DefaultConv2D(filters=128),
    DefaultConv2D(filters=128),
    tf.keras.layers.MaxPool2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(units=128, activation="relu",
kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(units=64, activation="relu",
kernel_initializer="he_normal"),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(units=1, activation="sigmoid")
])
```

```
model.compile(loss="binary_crossentropy", optimizer="nadam",  
              metrics=["accuracy"])  
model.summary()
```

Pretrained Models for Transfer Learning

```
train_set_raw=tf.data.Dataset.from_tensor_slices((X_train,y_train))  
valid_set_raw=tf.data.Dataset.from_tensor_slices((X_val,y_val))  
test_set_raw=tf.data.Dataset.from_tensor_slices((X_test,y_test))
```

```
tf.keras.backend.clear_session() # extra code – resets layer name counter  
batch_size = 32  
preprocess = tf.keras.applications.xception.preprocess_input  
train_set = train_set_raw.map(lambda X, y: (preprocess(tf.cast(X, tf.float32)),  
y))  
train_set = train_set.shuffle(1000, seed=42).batch(batch_size).prefetch(1)  
valid_set = valid_set_raw.map(lambda X, y: (preprocess(tf.cast(X, tf.float32)),  
y)).batch(batch_size)  
test_set = test_set_raw.map(lambda X, y: (preprocess(tf.cast(X, tf.float32)),  
y)).batch(batch_size)
```

```
# extra code – displays the first 9 images in the first batch of valid_set  
plt.figure(figsize=(12, 12))
```

```

for X_batch, y_batch in valid_set.take(1):
    for index in range(9):
        plt.subplot(3, 3, index + 1)
        plt.imshow((X_batch[index] + 1) / 2) # rescale to 0–1 for imshow()
        if(y_batch[index]==1):
            classt='FAKE'
        else:
            classt='REAL'
        plt.title(f"Class: {classt}")
        plt.axis("off")
plt.show()

```

```

data_augmentation = tf.keras.Sequential([
    tf.keras.layers.RandomFlip(mode="horizontal", seed=42),
    tf.keras.layers.RandomRotation(factor=0.05, seed=42),
    tf.keras.layers.RandomContrast(factor=0.2, seed=42)
])

```

extra code – displays the same first 9 images, after augmentation

```

plt.figure(figsize=(12, 12))
for X_batch, y_batch in valid_set.take(1):
    X_batch_augmented = data_augmentation(X_batch, training=True)
    for index in range(9):

```

```

plt.subplot(3, 3, index + 1)
# We must rescale the images to the 0-1 range for imshow(), and also
# clip the result to that range, because data augmentation may
# make some values go out of bounds (e.g., RandomContrast in this case).
plt.imshow(np.clip((X_batch_augmented[index] + 1) / 2, 0, 1))
if(y_batch[index]==1):
    classt='FAKE'
else:
    classt='REAL'
plt.title(f"Class: {classt}")
plt.axis("off")
plt.show()
tf.random.set_seed(42) # extra code – ensures reproducibility
base_model = tf.keras.applications.xception.Xception(weights="imagenet",
                                                    include_top=False)
avg = tf.keras.layers.GlobalAveragePooling2D()(base_model.output)
output = tf.keras.layers.Dense(1, activation="sigmoid")(avg)
model = tf.keras.Model(inputs=base_model.input, outputs=output)

for layer in base_model.layers:
    layer.trainable = False

optimizer = tf.keras.optimizers.SGD(learning_rate=0.1, momentum=0.9)
model.compile(loss="binary_crossentropy", optimizer=optimizer,

```

```

        metrics=["accuracy"])
history = model.fit(train_set, validation_data=valid_set, epochs=3)

for indices in zip(range(33), range(33, 66), range(66, 99), range(99, 132)):
    for idx in indices:
        print(f"{idx:3}: {base_model.layers[idx].name:22}", end="")
    print()

model.evaluate(test_set)

for layer in base_model.layers[56:]:
    layer.trainable = True
    optimizer = tf.keras.optimizers.SGD(learning_rate=0.01, momentum=0.9)
    model.compile(loss="binary_crossentropy", optimizer=optimizer,
        metrics=["accuracy"])
    history = model.fit(train_set, validation_data=valid_set, epochs=10)

# plot model performance
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']

```

```

loss = history.history['loss']
val_loss = history.history['val_loss']
epochs_range = range(1, len(history.epoch) + 1)
plt.figure(figsize=(15,5))
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Train Set')
plt.plot(epochs_range, val_acc, label='Val Set')
plt.legend(loc="best")
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Model Accuracy')
plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Train Set')
plt.plot(epochs_range, val_loss, label='Val Set')
plt.legend(loc="best")
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Model Loss')
plt.tight_layout()
plt.show()
model.evaluate(test_set)

```

Add Explainability to the model

```
!pip install lime
```

```
from lime import lime_image
```

```

explainer = lime_image.LimeImageExplainer()

plt.figure(figsize=(12, 12))
for index in range(9):
    plt.subplot(3, 3, index + 1)
    plt.imshow((x[index] + 1) / 2) # rescale to 0–1 for imshow()
    if(y[index]==1):
        classt='FAKE'
    else:
        classt='REAL'
    plt.title(f"Class: {classt}")
    plt.axis("off")
plt.show()

```

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